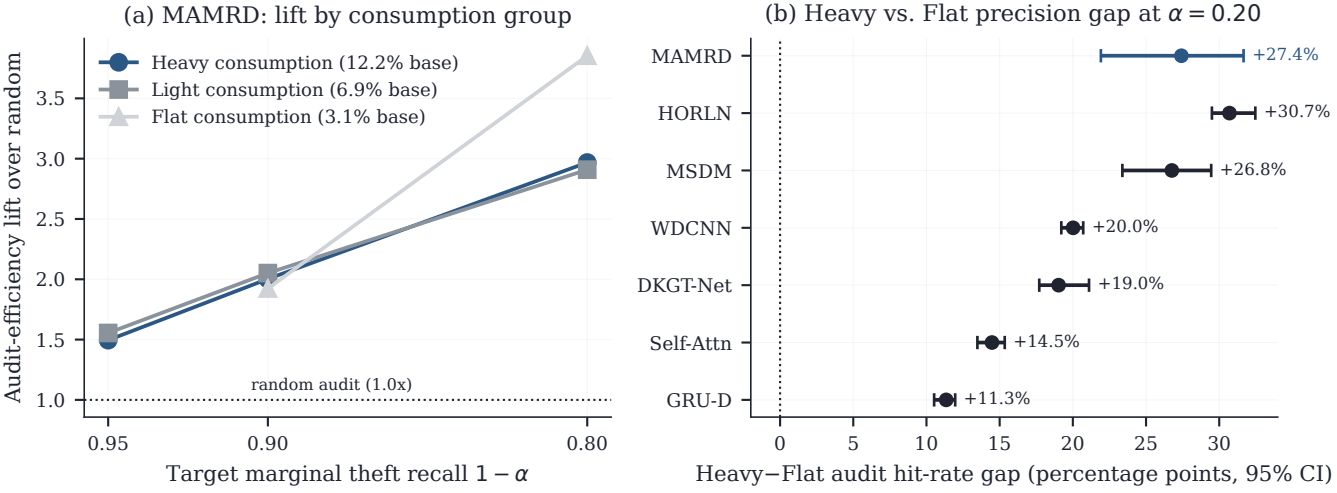
(a) MAMRD: lift by consumption group
Heavy consumption (12.2% base)
Light consumption (6.9% base)
Flat consumption (3.1% base)
Audit-efficiency lift over random
3.5
3.0
2.5
2.0
1.5
1.0
random audit (1.0x)
0.95
0.90
0.80
Target marginal theft recall 1 − α
(b) Heavy vs. Flat precision gap at α = 0.20
MAMRD
HORLN
MSDM
WDCNN
DKGT-Net
Self-Attn
GRU-D
+27.4%
+30.7%
+26.8%
+20.0%
+19.0%
+14.5%
+11.3%
0
5
10
15
20
25
30
Heavy−Flat audit hit-rate gap (percentage points, 95% CI)